

Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions



Dataset Overview

3,900

Total Purchases

Transactions analyzed
across all categories

18

Data Columns

Comprehensive
customer and
purchase attributes

50

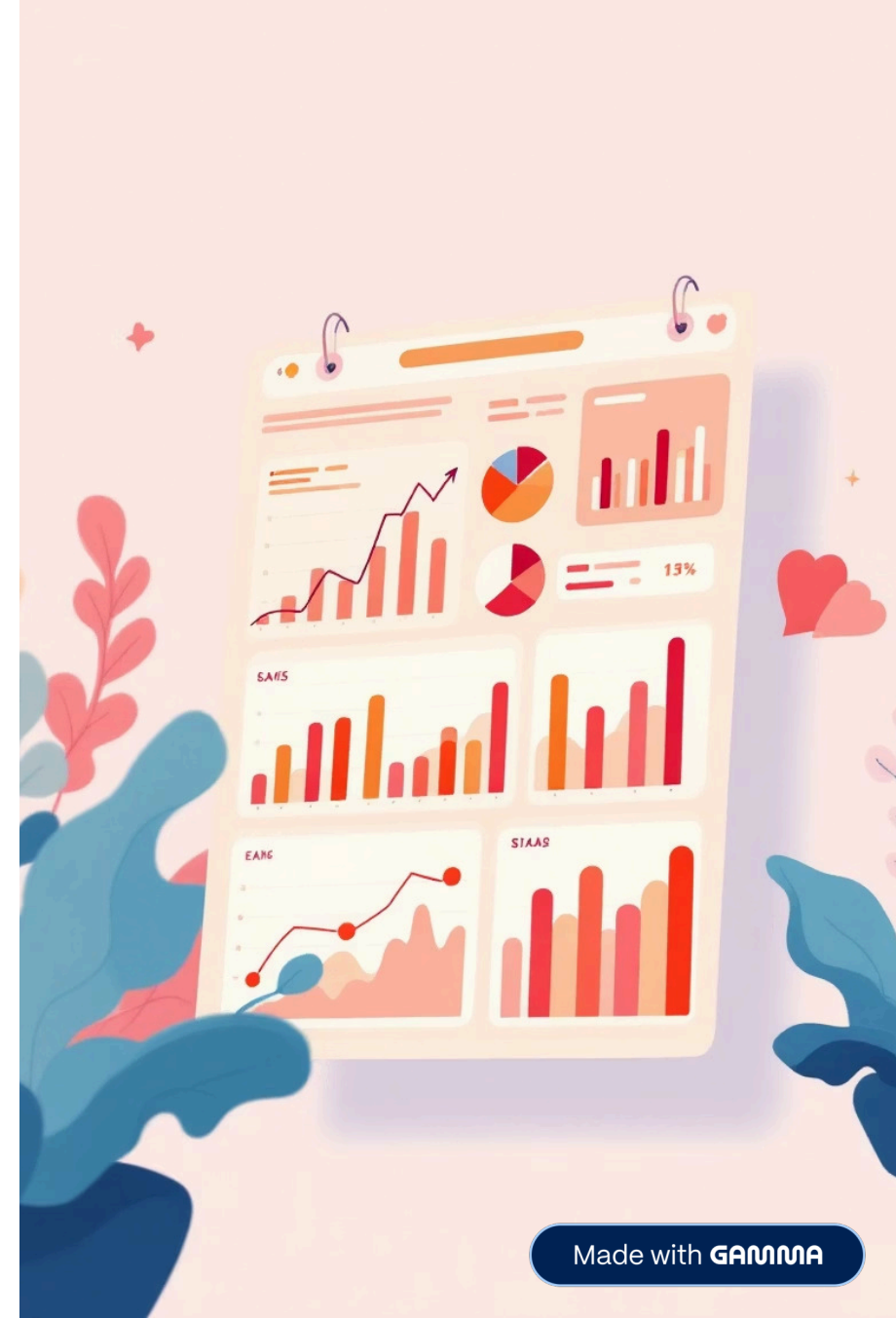
Locations

Geographic diversity in
customer base

25

Product Types

Wide variety of items
purchased





Data Preparation Journey

01

Data Loading & Exploration

Imported dataset using pandas, analyzed structure with `df.info()` and summary statistics

02

Missing Data Handling

Imputed 37 missing Review Rating values using median rating per product category

03

Feature Engineering

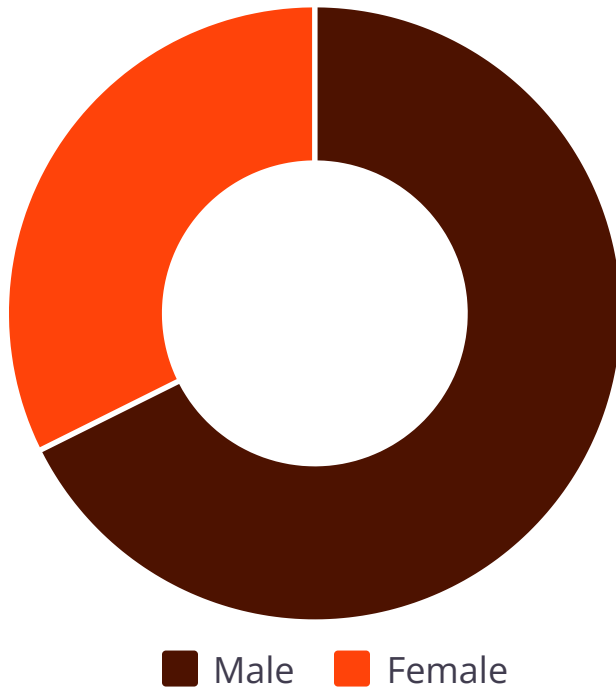
Created `age_group` and `purchase_frequency_days` columns for deeper insights

04

Database Integration

Connected to PostgreSQL and loaded cleaned data for SQL analysis

Revenue by Gender



Key Insight

Male customers generate 68% of total revenue, significantly outpacing female customers

Strategic opportunity to boost female customer engagement

Top-Rated Products

Gloves

★ 3.86 rating

Sandals

★ 3.84 rating

Boots

★ 3.82 rating

Hat

★ 3.80 rating

Skirt

★ 3.78 rating

Highlight these top-rated products in marketing campaigns to drive sales



Shipping & Spending Patterns

Express Shipping

\$60.48

Average purchase amount

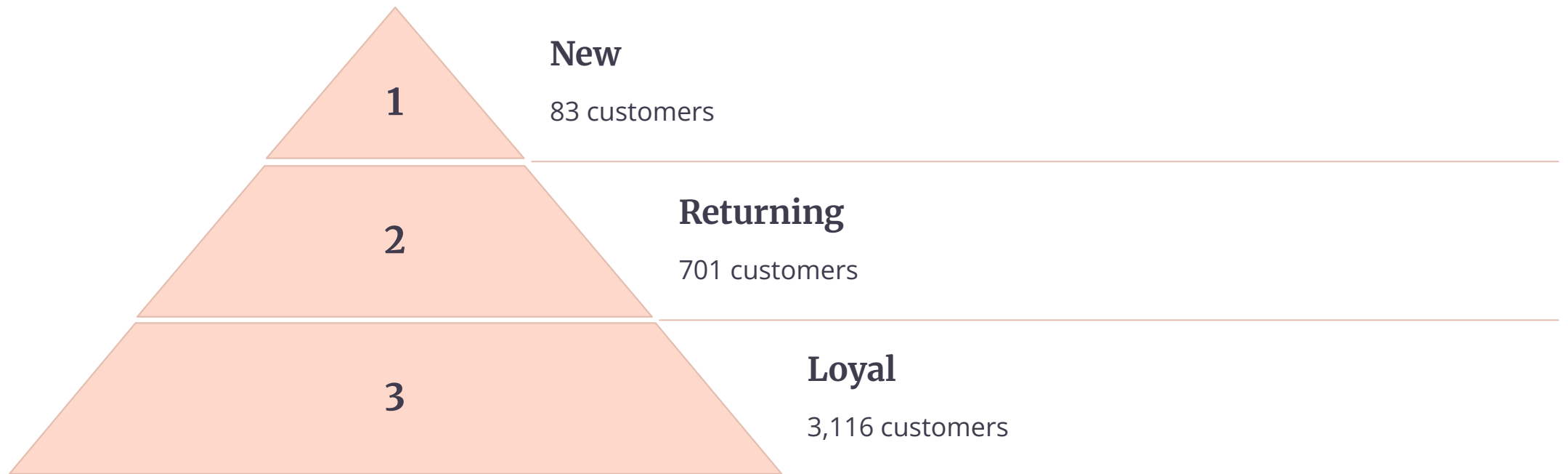
Standard Shipping

\$58.46

Average purchase amount

Express shipping customers spend 3.5% more per transaction

Customer Segmentation Analysis



Subscription Status

- 1,053 subscribers (27%)
- 2,847 non-subscribers (73%)
- 958 repeat buyers with subscriptions

Revenue Impact

Subscribers: **\$62,645** total revenue

Non-subscribers: **\$170,436** total revenue

Similar avg spend: ~\$59-60 per purchase

Discount Strategy Insights

High-Value Discount Users

839 customers used discounts but spent above average (\$59.76)

Discount-Dependent Products

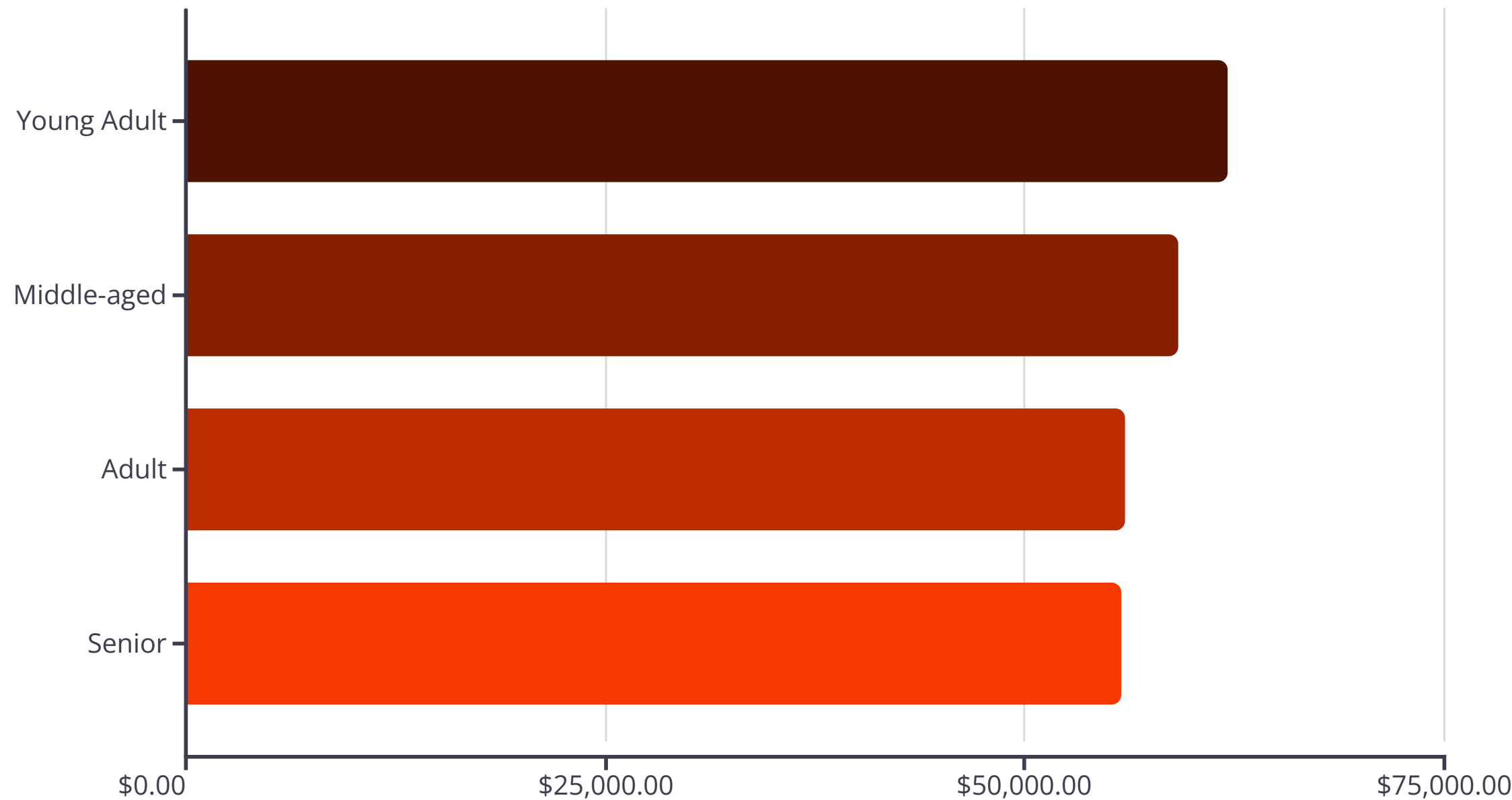
Hat (50%), Sneakers (49.66%), Coat (49.07%) have highest discount rates

Strategic Opportunity

Balance promotional sales with margin control to optimize profitability



Revenue by Age Group



Young adults lead revenue generation, followed closely by middle-aged customers

Strategic Recommendations



Boost Subscriptions

Promote exclusive benefits to convert non-subscribers



Loyalty Programs

Reward repeat buyers to expand loyal segment



Review Discount Policy

Balance sales boosts with margin control



Product Positioning

Highlight top-rated items in campaigns



Targeted Marketing

Focus on high-revenue age groups and express shipping users



Thank You